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CSD 380 – DevOps

Module 8.2 Assignment – The Dangers of Change Approval Processes

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Why Traditional Approval Processes Undermine DevOps Velocity

Introduction

Change approval processes are commonly used in organizations to manage modifications to systems, projects, and infrastructure. Typically, these processes involve documenting a proposed change, submitting it to a review committee, and obtaining formal authorization before the change is implemented. Traditional IT frameworks, such as ITIL, have emphasized the importance of structured change approval. While these processes were designed to mitigate risks, they can have negative consequences if they become too rigid or overly controlled.

In modern DevOps environments, the drawbacks of manual change approval have become increasingly evident. This paper examines the risks associated with stringent change approval procedures and discusses why organizations should assess how these controls impact their velocity, reliability, and ability to innovate.

This paper examines the risks associated with strict change approval procedures. It explains why organizations should assess how these controls impact their speed, reliability, and capacity to innovate, promoting a transition toward automated and risk-based change validation.

Delays and Reduced Delivery Speed

One of the most cited dangers of formal change approval processes is the potential for delays. When every change must pass through a centralized group, such as a Change Advisory

Board (CAB), it can slow down the pace of work. Research on software delivery consistently shows that external approval bodies often increase lead times and reduce deployment frequency (Forsgren, 2021). These delays can cause organizations to fall behind their competitors, extend project timelines, and raise operational costs (Smart Approve, 2023). When large groups or committees are required to weigh in on routine changes, the workflow becomes reliant on meeting schedules and manual reviews rather than on technical readiness.

Decreased Agility and Innovation

Overly strict change approval processes can hinder an organization's ability to adapt. When developers or administrators realize that even minor changes will necessitate time-consuming paperwork and reviews, they may be reluctant to propose improvements. This discouragement stifles innovation and prevents employees from experimenting or introducing enhancements (LaunchDarkly, 2023).

Modern DevOps culture prioritizes continuous improvement, short feedback cycles, and rapid iteration. However, heavy approval processes conflict with these objectives by subjecting all changes, regardless of their risk level, to the same rigid steps. As a result, organizations often miss valuable opportunities to improve their systems quickly.

Increased Failure Risk and Workarounds

Although approval processes are meant to enhance safety, research shows that manual approvals do not significantly improve the success rates of changes (Forsgren, 2021). In many instances, the added bureaucracy leads teams to find ways to bypass formal procedures. These workarounds might include undocumented changes, unapproved schedules, or last-minute fixes implemented outside of the established process. Such actions increase the risk of outages since

they avoid necessary testing and oversight. Additionally, excessive approval requirements may promote “big bang” releases, where teams combine multiple changes into a single deployment to reduce interactions with the approval board. Larger releases result in more complex failure scenarios, making troubleshooting more challenging.

Resource Waste and Organizational Inefficiency

Complex change approval procedures require a significant amount of time and effort. Engineers must prepare detailed documentation, managers need to review and approve changes, and committees must convene to discuss them. This process consumes hours that could otherwise be dedicated to development, strategic planning, testing automation, or other productive activities. Additionally, the slow pace of approvals results in indirect costs, including extended project durations and increased administrative overhead. In many cases, the cost of maintaining an extensive approval framework outweighs its benefits.

Negative Impact on Team Morale

Strict or bureaucratic change approval processes often frustrate technical staff. Developers and engineers may see repetitive documentation requirements as unnecessary obstacles. When approval bodies deny or delay changes without clear justification, it creates tension between teams. Over time, this can lower morale and contribute to burnout or turnover. Modern high-performing teams expect a degree of autonomy and trust, especially as automated testing and monitoring reduce the need for manual oversight. An inflexible approval structure can harm the culture necessary for the successful adoption of DevOps.

False Sense of Security

One of the risks associated with traditional approval processes is the false sense of safety they create. Approving a change does not ensure that it has been adequately tested, integrated, or aligned with technical requirements. When organizations rely too much on formal approvals instead of implementing automated validation and continuous monitoring, they miss out on opportunities to identify real risks. Executives may mistakenly believe that having a documented approval process is enough for compliance, even though this process might not actually prevent misconfigurations or failures. This misplaced confidence can lead to significant operational problems.

Conclusion

Change approval processes are essential for governance, compliance, and coordination. However, when these processes are overly restrictive or manual, they can pose significant risks. Delays, reduced agility, increased chances of failure, wasted resources, low morale, and a false sense of security can all harm organizational performance. In DevOps environments, where speed and reliability are paramount, organizations should reassess their traditional change approval structures and adopt risk-based, automated, and streamlined models. This approach enables teams to maintain safety and compliance through automated validation, ultimately preserving the speed and flexibility needed to deliver high-quality systems quickly and consistently in a modern DevOps setting.

References

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